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Supercapacitor-Powered BICOLOUR LED NIGHT LAMP

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You can make a bicolour LED night lamp powered by a supercapacitor that is based on joule thief circuit for charging the supercapacitor. Most night lamps go off when AC mains power fails. This night lamp works even when there is power outage. When fully charged, the supercapacitor can drive the LED light for long hours and even days.

The lamp's light can be green or red, or even white in colour. It can also be used as an emergency lamp. LED night lamp is safer and more energy-efficient than AC mains operated incandescent bulb or tubelight night lamps.

Circuit and working

The circuit diagram of supercapacitor LED night lamp is shown in Fig. 2. It has transistors T1 (2N3904) and T2 (BC548), a toroidal transformer with coils L1 and L2, 2.7V/100F supercapacitor C1, thirteen LEDs (LED1 through LED13), Zener diode ZD1, and a few other components.

When the circuit is powered by 12V DC, only green LEDs (LED5 through LED12) glow. Yellow LED13 glows only when the supercapacitor is fully charged. When the power supply goes off, red LEDs (LED1 through LED4) glow.

The forward voltage of a red LED is about 2V only, unlike other LEDs like white or blue for which it is about 3V or more. So red LEDs are used here as they glow for longer time as compared to other LEDs. The author's prototype uses four red LEDs which glow for six days with a fully charged capacitor. You can replace some or all the LEDs with white LEDs for brighter light but then you will have to charge the capacitor more often. Switch S1 is provided to switch off the lamp when it is not to be used.

Joule thief circuit is a low-cost and easy-to-build self-oscillating voltage booster that is widely used for driving small loads like LED lights. Its main components include a toroidal transformer, two transistors, a supercapacitor, and four red LEDs (LED1 through LED4). If we directly connect LEDs to the supercapacitor (Fig. 3), the LEDs will be on till the capacitor voltage drops to 2V (Vf). After that the stored energy remains unused.

Fig. 1: Author's prototype

Joule thief circuit here makes use of NPN transistor T1 and ferrite core transformer with coils L1 and L2. These coils are connected to capacitor voltage Vc. The other end of coil L1 is connected to collector of transistor T1 and coil L2 to its base through current-limiting resistor R3.

The current in coil L1 builds up and induces voltage in coil L2, driving transistor T2. Once the current in L1 stops increasing, there is no rate of change of current in it, so no voltage is induced in L2. This turns off transistor T1. The stored energy in L1 produces a large voltage spike and drives LEDs LED1 through LED4 that are connected to collector of T1. Once the stored energy in L1 is consumed by the LEDs, transistor T1 again turns on and the cycle repeats.

Construction and testing

An actual-size PCB layout for the supercapacitor LED night lamp is shown in Fig. 4 and its component layout in Fig. 5. Supercapacitor C1 can be either mounted on the PCB or directly connected with two wires to the positive and negative terminals and placed suitably inside the cabinet.

The four red LEDs are powered by joule thief circuit. In the author's prototype, all the LEDs

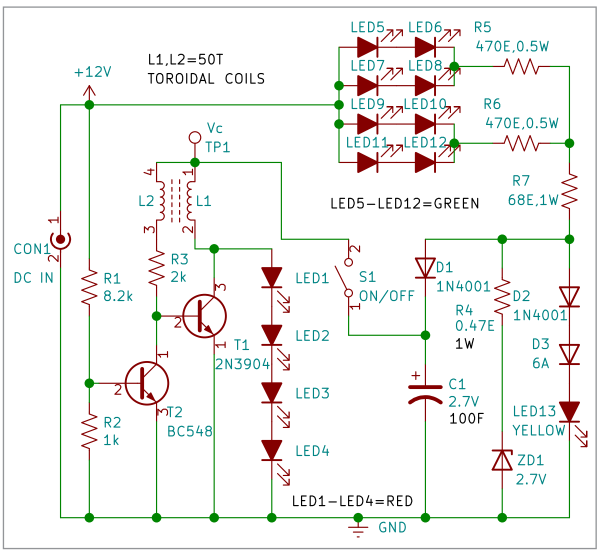

Fig. 2: Circuit diagram

Fig. 3: Supercapacitor used in the project